Local and global well-posedness of the scalar Brazovskii gradient flow

TECT verification-first repository · claim A2-PDE-WELLPOSED

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1. Purpose and scope

This note establishes the well-posedness foundation A2 for the scalar Brazovskii gradient flow, on which the Sector-B free-energy minimisation rests dynamically. The argument is standard semilinear-parabolic theory (Henry 1981; Pazy 1983; Lunardi 1995). **Scope (deliberately narrow, operator review 2026-06-23):** the result proved here is

$$\mathrm{A2}_{\text{scalar, periodic, } \mu^2>0,\ \gamma>0}: \quad \tfrac{3}{2} < s \le 2 \implies \text{global well-posedness,} \tag{1}$$

i.e. ONE real scalar field on a FIXED periodic cell with $\mu^2 > 0$, $\gamma > 0$. It does **not** cover the full Class-II multi-field action, nor the $\mu^2 < 0$ condensate branch of the numerical mainline; those are separate proof targets ($\mathrm{A2}_{\text{full Class II}}$, $\mathrm{A2}_{\mu^2<0}$). In particular $\mu^2 > 0$ is the *disordered-side* condition giving a positive linear gap; it is *not* a proof of BCC condensation. Numerical constants are pinned by `codes/foundations/a2_wellposedness_checks.py` (8/8 PASS).

2. Setup

Let $\mathbb{T}^3 = \mathbb{R}^3/\Lambda$ be the fixed periodic cell, Λ^* its reciprocal lattice. With the A1-KERNEL-CONV kernel $K(q) = \mu^2 + Y(q^2 - q_0^2)^2$ ($Y > 0$),

$$F_{\mathrm{TECT}}[\phi] = \int_{\mathbb{T}^3} \Big[\tfrac{1}{2}\phi K(-i\nabla)\phi + \tfrac{\lambda}{4}\phi^4 + \tfrac{\gamma}{6}\phi^6\Big] dx, \tag{2}$$

with L^2 gradient flow, on $X := L^2(\mathbb{T}^3)$,

$$\partial_t \phi = -L\phi + N(\phi), \quad L := K(-i\nabla), \quad N(\phi) := -\lambda\phi^3 - \gamma\phi^5, \quad \phi(0) = \phi_0. \tag{3}$$

Standing hypotheses **(H1)** $\mu^2 > 0$, **(H2)** $\gamma > 0$ (production anchor $\mu^2 = 5 \times 10^{-3}$, $\gamma = 1.62$).

3. The linear part generates an analytic semigroup

L is the self-adjoint Fourier multiplier $K(k)$, $k \in \Lambda^*$, with $D(L) = H^4$. The relevant spectral fact is the *lower bound*

$$\lambda_0 := \min_{k\in\Lambda^*} K(k) \ \ge\ \mu^2 \ >\ 0 \qquad \text{(H1)}, \tag{4}$$

since $K(k) = \mu^2 + Y(\cdots)^2 \ge \mu^2$ for every k. Equality $\lambda_0 = \mu^2$ holds only if some lattice mode lies exactly on the shell $|k| = q_0$; on a generic cell $\lambda_0 > \mu^2$ (assert `lambda0_lower_bound_mu2_positive`: μ^2 is the sharp continuum infimum, lattice $\lambda_0 > \mu^2$). Only $\lambda_0 > 0$ is used: a positive self-adjoint operator is sectorial, so $-L$ generates an analytic semigroup e^{-tL}, fractional powers L^α exist with $D(L^\alpha) = H^{4\alpha} =: X^\alpha$, and

$$\|L^\beta e^{-tL}\|_{X\to X} \le C_\beta t^{-\beta}, \quad C_\beta = \sup_{x\ge 0} x^\beta e^{-x} = \beta^\beta e^{-\beta} < 1 \quad (0 < \beta < 1). \tag{5}$$

Moreover $K(k) \asymp (1 + |k|^2)^2$ with constants depending on μ^2, Y, q_0 (lower constant positive by (4)), so $\langle \phi, L\phi\rangle$ is equivalent to $\|\phi\|_{H^2}^2$.

4. The nonlinearity is locally Lipschitz on X^β

Fix $\beta \in (\frac{3}{8}, \frac{1}{2}]$, so $X^\beta = H^{4\beta}$ with $\frac{3}{2} < 4\beta \le 2$. By Sobolev in $d = 3$, $H^{4\beta} \hookrightarrow L^\infty$ for $4\beta > \frac{3}{2}$ (assert `sobolev_linf_threshold`), and $H^{4\beta}$ is a Banach algebra. Hence on balls $\{\|\cdot\|_{X^\beta} \le R\}$,

$$\|N(\phi) - N(\psi)\|_X \le \ell(R)\|\phi - \psi\|_{X^\beta}, \qquad \ell(R) = C\,(|\lambda|R^2 + \gamma R^4), \tag{6}$$

so $N : X^\beta \to X$ is locally Lipschitz and bounded on bounded sets.

5. Local existence, uniqueness, regularity ($\frac{3}{2} < s \le 2$)

With L sectorial (§3) and $N : X^\beta \to X$ locally Lipschitz (§4), Henry (1981, Thm. 3.3.3) gives, for every $\phi_0 \in X^\beta = H^{4\beta}$ ($s := 4\beta \in (\frac{3}{2}, 2]$), a unique mild solution

$$\phi \in C([0,\tau]; H^s) \cap C((0,\tau]; H^4) \cap C^1((0,\tau]; X), \quad \phi(t) = e^{-tL}\phi_0 + \int_0^t e^{-(t-\sigma)L} N(\phi(\sigma))d\sigma, \tag{7}$$

the integral converging by (5). Uniqueness and continuous dependence follow by Gronwall; parabolic smoothing plus smoothness of N bootstrap $\phi \in C^\infty((0,\tau) \times \mathbb{T}^3)$. **The range is $\frac{3}{2} < s \le 2$:** the local theory used here employs only X^β, $\beta \le \frac{1}{2}$. Persistence in H^s for $s > 2$ requires a separate higher-order estimate not carried out here and is therefore not claimed.

6. Global existence ($\frac{3}{2} < s \le 2$)

F_{TECT} is a strict Lyapunov functional: $\frac{d}{dt}F[\phi(t)] = -\|\delta F/\delta\phi\|_{L^2}^2 \le 0$. Under (H2), by Young $\frac{|\lambda|}{4}t^4 \le \frac{\gamma}{12}t^6 + C_*$ ($C_* = 1.01 \times 10^{-2}$, assert `sextic_dominates_quartic_coercive`),

$$F_{\text{TECT}}[\phi] \ge \tfrac{1}{2}\langle \phi, L\phi\rangle + \tfrac{\gamma}{12}\|\phi\|_{L^6}^6 - C_*|\mathbb{T}^3|, \tag{8}$$

so $F[\phi(t)] \le F[\phi_0]$ gives an a priori bound in H^2 (§3 norm equivalence). **Inclusion direction (corrected):** since $\beta \le \frac{1}{2}$, i.e. $4\beta \le 2$, one has $H^2 \subseteq H^{4\beta} = X^\beta$ (the smoother space embeds in the less smooth one), hence $\|\phi\|_{X^\beta} \le C\|\phi\|_{H^2}$ and the H^2 a priori bound *controls* the X^β norm (assert `continuation_window_nonempty`). The continuation alternative then forces $\tau = \infty$.

Theorem (A2, scalar/periodic/$\mu^2 > 0/\gamma > 0$). *Under (H1) and (H2), for every $\phi_0 \in H^s(\mathbb{T}^3)$ with $\frac{3}{2} < s \le 2$ the flow (3) has a unique global solution $\phi \in C([0,\infty); H^s) \cap C^\infty((0,\infty) \times \mathbb{T}^3)$, depending continuously on ϕ_0, with $F_{\text{TECT}}[\phi(t)]$ non-increasing.*

7. Devil's-advocate (operator review 2026-06-23)

(α) *"$\inf\operatorname{spec} L = \mu^2$ is false on a generic cell."* UPHELD and fixed: the proof now uses only the lower bound $\lambda_0 := \min_k K(k) \ge \mu^2 > 0$ (4); equality is not claimed (it is non-generic). Sectoriality and the H^2 equivalence need only $\lambda_0 > 0$.

(β) *"The Sobolev inclusion is backwards."* UPHELD and fixed: for $\beta \le \frac{1}{2}$ the correct direction is $H^2 \subseteq H^{4\beta} = X^\beta$ (§6), which still yields "H^2 bound controls X^β" — the conclusion survives in the correct direction.

(γ) *"The proof only closes for $\frac{3}{2} < s \le 2$."* UPHELD: the theorem range is now stated as $\frac{3}{2} < s \le 2$; $s > 2$ persistence is flagged as a separate higher-order estimate, not claimed. Scope narrowed to one scalar field on a fixed periodic cell, $\mu^2 > 0$ (disordered side, NOT BCC condensation), $\gamma > 0$; full Class-II and $\mu^2 < 0$ remain separate targets.

Quantitative sanity check (mandatory): all constants reproduced by the 8/8 verification (v1.1): $\lambda_0 \geq \mu^2 = 5 \times 10^{-3}$ with strict $\lambda_0 > \mu^2$ on a generic lattice; Sobolev $s > 3/2$, $\beta > 3/8$; smoothing $C_\beta = \beta^\beta e^{-\beta} \in (0.43, 0.48)$; Young $C_* = 1.01 \times 10^{-2}$; continuation window $\beta \in (3/8, 1/2]$, $4\beta \leq 2$, $H^2 \subseteq X^\beta$.

8. Result footer

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Result ID:                A2-PDE-WELLPOSED (scalar, periodic, mu^2>0, gamma>0)
Precise statement:        Under mu^2>0 and gamma>0, the scalar Brazovskii L2
                          gradient flow on a fixed T^3 is globally well-posed for
                          phi_0 in H^s, 3/2 < s <= 2: unique global solution,
                          smooth for t>0, continuous dependence, F non-increasing.
Scope:                    ONE real scalar field, fixed periodic cell, 3/2<s<=2.
                          mu^2>0 = disordered-side positive linear gap (NOT BCC
                          condensation). NOT full Class-II; NOT mu^2<0 branch.
Dependencies:             A1-KERNEL-CONV.
Hypotheses:               (H1) mu^2>0 [lambda0=min_k K(k)>=mu^2>0: sectoriality +
                          H^2 equivalence]; (H2) gamma>0 [coercivity, global].
Evidence grade:           PROVED CONDITIONAL (analytic) + EXECUTED (8/8 checks).
Reproduction command:     python codes/foundations/a2_wellposedness_checks.py
Expected output:          8/8 PASS; exit 0.
Falsification gate:       finite-time blow-up or non-uniqueness for some phi_0 in
                          H^s (3/2<s<=2) at mu^2>0, gamma>0; or lambda0:=min_k
                          K(k) <= 0.
Tier before / after:      A2 T1 / T6 PROVED CONDITIONAL (scalar/periodic/mu^2>0
                          scope) -- RE-SUBMITTED for operator sign-off after the
                          2026-06-23 review corrections.
No-overclaim statement:   No s>2 persistence; no full Class-II; no mu^2<0
                          condensate branch; mu^2>0 is NOT BCC condensation; no
                          unique global minimiser (that is Sector B).
Next required action:     Operator sign-off of the narrowed T6; then A3
                          (renormalisation / continuum limit); separately
                          A2_full-ClassII and A2_{mu^2<0}.
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